Examiner only

2. Diagrams **A**, **B**, **C** and **D** represent hydrogen (H_2), sulfur dioxide (SO_2), hypochlorous acid (HClO) and phosphine (PH_3) but **not in that order**.

A **B** **C** **D**

(a) (i) Give the **letter** of the diagram that represents an element. Give a reason for your answer. [2]

Letter ....

Reason ....

....

(ii) Give the **letter** of the diagram that represents SO_2. [1]

....

(iii) Use diagrams **A**, **B**, **C** and **D** above to work out which of the following diagrams represents water (H_2O). **Circle** the correct diagram. [1]

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(b) (i) Calculate the relative molecular mass (M_r) of hypochlorous acid, HClO. [1]

A_r(H) = 1 A_r(O) = 16 A_r(Cl) = 35.5

M_r = ...

(ii) The relative molecular mass (M_r) of sulfur dioxide is 64.

Calculate the percentage by mass of sulfur in sulfur dioxide, SO_2. [2]

A_r(S) = 32

Percentage = ... %

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(c) Phosphine (PH_3) is made when phosphorus and hydrogen react.

Put a tick (✓) in the box next to the correctly balanced symbol equation for this reaction. [1]

$2P + 3H_2 \longrightarrow PH_3$	☐
$2P + 3H_2 \longrightarrow 2PH_3$	☐
$2P + 3H_2 \longrightarrow 3PH_3$	☐
$2P + 3H_2 \longrightarrow 4PH_3$	☐

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